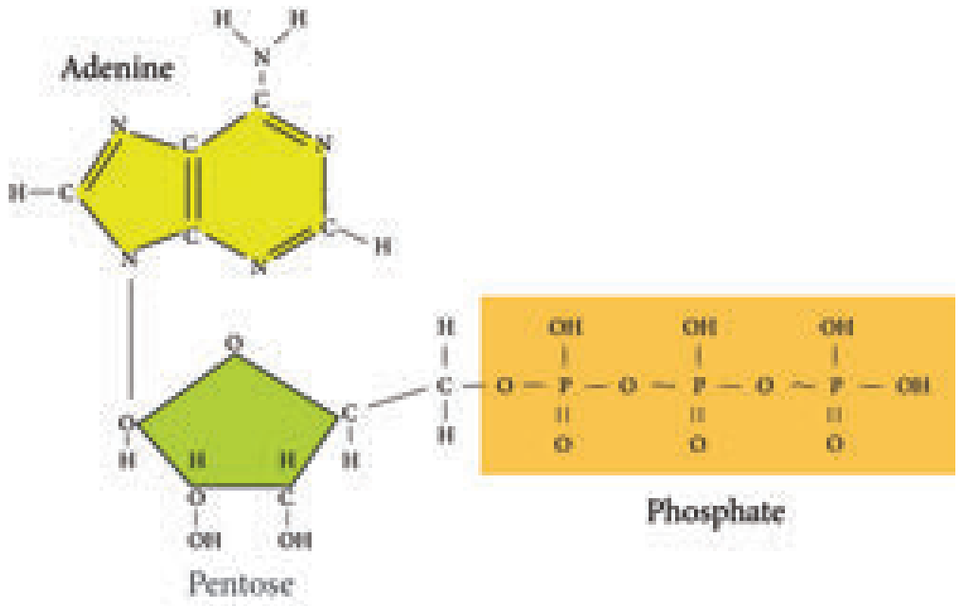


Figure 4.2: Structure of ATP

4.2.2 Oxidation – Reduction Reactions

Oxidation is the removal of electrons (e^-) from an atom or molecule and is often an energy producing reaction. Reduction of a substrate refers to its gain or addition of one or more electrons to an atom or molecule. Oxidations and reduction are always coupled. In other words, each time one substance is oxidized, another is simultaneously reduced.

$F_2 + 2e^- \longrightarrow 2F^-$

$H_2 + 2e^- \longrightarrow 2H^+ + 2e^-$

$NAD^+ + 2H^+ + 2e^- \rightleftharpoons NADH + H^+$

4.3 Generation of ATP

Much of energy released during oxidation reduction reaction is trapped within the cell by the formation of ATP. A phosphate group is added ADP with the input of energy to form ATP. The addition of a phosphate to a chemical compound is called phosphorylation.

Organism uses three different mechanisms of phosphorylation to generate ATP from ADP. They are

4.3.1 Substrate Level Phosphorylation

It is a metabolic reaction that results in the formation of ATP or GTP by the direct transfer of a phosphoryl group to ADP or GDP from another phosphorylated compound.



Alkaline phosphatase is a heat sensitive enzyme in milk which is used as an indicator in Pasteurization.

4.3.2 Oxidative Phosphorylation

In this reaction, electrons are transferred from organic compounds to molecules of Oxygen (O_2) or other inorganic molecules through a series of different electron carriers (Example: NAD^+ and FAD. Then the electrons are passed through a series of different electron carriers to molecules of oxygen. The process of oxidative phosphorylation occurs during electron transport chain (Figure 4.3).

TN_GOVT_XII_Micro_Biology_CH04.indd 48 04-03-2019 12:54:55